

SINR-Aware Deep Reinforcement Learning for Distributed Dynamic Channel Allocation in Cognitive Interference Networks

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Abstract—We consider the problem of dynamic channel allocation (DCA) in cognitive communication networks with the goal of maximizing a global signal-to-interference-plus-noise ratio (SINR) measure under a specified target quality of service (QoS)-SINR for each network. The shared bandwidth is partitioned into K channels with frequency separation. In contrast to the majority of existing studies that assume perfect orthogonality or a one-to-one user-channel allocation mapping, this paper focuses on practical systems experiencing inter-carrier interference (ICI) and channel reuse by multiple large-scale networks. This realistic scenario significantly increases the problem dimension, rendering existing algorithms inefficient. We propose a novel multi-agent reinforcement learning (RL) framework for distributed DCA, named Channel Allocation RL To Overlapped Networks (CARLTON). The CARLTON framework is based on the Centralized Training with Decentralized Execution (CTDE) paradigm, utilizing the DeepMellow value-based RL algorithm. To ensure robust performance in the interference-laden environment we address, CARLTON employs a low-dimensional representation of observations, generating a QoS-type measure while maximizing a global SINR measure and ensuring the target QoS-SINR for each network. Our results demonstrate exceptional performance and robust generalization, showcasing superior efficiency compared to alternative state-of-the-art methods, while achieving a marginally diminished performance relative to a fully centralized approach.

Index Terms – Distributed dynamic channel allocation (DCA), wireless communications, cognitive interference networks, signal-to-interference-plus-noise ratio (SINR), deep reinforcement learning (DRL).

I. INTRODUCTION

The growing need for wireless communications, alongside the challenges posed by spectrum scarcity, has prompted the advancement of effective Dynamic Channel Allocation (DCA) schemes for emerging cognitive communication network technologies. Our focus is on DCA within large-scale networks, aiming to optimize the allocation of frequency channels among networks sharing the same frequency band. Unlike static channel allocation, where frequencies remain fixed, DCA adapts to changing network conditions by dynamically allocating channels based on factors such as interference and signal strength, impacting user rates in the network. DCA enhances the efficiency and flexibility of wireless networks, ensuring efficient channel utilization and mitigating the impact of interference, leading to improved overall performance and quality

of service (QoS). It offers advantages in common communication environments, including scalability, robustness, and lower latency. It allows each node to make real-time decisions based on local information, enabling fast convergence in dynamic environments. However, challenges such as coordination and interference management, as well as the complexity of algorithm design, must be addressed to ensure high performance. Recent studies on this topic, which involves learning the environment and channel allocation algorithms, have employed frameworks such as multi-armed bandits (MAB) [1]–[9], stable matching [10], [11], game theoretic optimization and congestion control [1], [8], [12]–[21], and, more recently, deep reinforcement learning (DRL) for multi-user scenarios [22]–[37]. The latter was initially explored in our earlier work (Naparstek and Cohen [22], [23]) within a multi-agent framework, following single-agent DRL research in [38], paving the way for a significant amount of subsequent research in the wireless communications community.

A. DRL Algorithms for DCA

DCA algorithms can be classified into two distinct branches: Centralized and distributed methods. Related studies within the centralized approach have utilized techniques such as graph vertex coloring, where adjacent vertices (representing users or local networks) are modeled with strong mutual interference, necessitating their allocation to different channels. Colors represent orthogonal channels, effectively coloring the graph. This paper addresses more challenging scenarios where interference exists between channels, and varies across the graph, for instance, decreasing with geographical distance, contrasting the binary interference assumption in graph coloring models. Additionally, centralized DRL methods have been recently proposed (see our recent work [35] and references therein). The limitation of the centralized approach lies in the communication overhead and computational complexity required for solving centralized DCA optimization. Consequently, in many applications, the preference is for distributed DCA, as it is widely considered more suitable for various wireless communication scenarios due to the intricate nature of the problem. For instance, in cognitive radio networks, distributed DCA allows for efficient spectrum utilization and adaptability to varying spectrum availability. Similarly, in IoT networks, distributed DCA supports scalable and efficient communications among a large number of devices.

In this paper, we focus on DRL methodology to address the distributed DCA problem. In recent years, the notable achievements of DRL in attaining strong performance surpassing human capabilities across various game domains have

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captured the attention of researchers from various disciplines. Prominent examples include its successes in games such as Atari [39], [40], Go [41], and even in complicated game such as StarCraft II [42]. This success has motivated researchers to explore the development of DRL in diverse fields such as robotics, biology, chemistry, finance [43], as well as wireless communications [22]–[31], [34], [44]. This approach is particularly promising in the topic of DCA, where the collection of data becomes inefficient due to the high dimensionality of the state space resulting from the numerous uncertainties inherent in the environment. Particularly, recent study [33] proposed a model-free Multi-Agent Reinforcement Learning (MARL) independent Q-learning (IQL) table approach to solve the distributed DCA problem over orthogonal channels. While this work demonstrates the potential of a simple RL technique for addressing the DCA challenge, it is limited to small-scale systems due to the high dimensionality of the state space in the DCA problem. To overcome this limitation, the state space was reduced and strongly quantized in the conducted study. Additionally, the IQL technique, both in a general context and specifically in the domain of radio systems, imposes constraints on system scalability. This arises from the inherent characteristic wherein each agent acquires an independent policy, which is greatly dependent on the policies formulated by the other participating agents. Furthermore, recently in [36], the authors employed the Double Deep Q-network (DDQN) [45] as the RL optimization mechanism to solve DCA in multi-agent systems. This utilization of the DDQN facilitates the exploration of more effective strategies. The transition from a tabular approach to neural networks (NN) diminishes the dependence on quantization techniques employed in previous research, leading to an improved generalization of the problem. In [32], an attention-based Graph Neural Network (GNN) architecture known as GA-Net, was introduced. The study addressed DCA in 6G sub-networks using a centralized training approach and a decentralized execution paradigm. The authors assumed channel orthogonality and employed a modified Soft-Actor-Critic (SAC) algorithm as the optimization technique for MARL system. By combining the hard attention layer with the Graph Attention Network (GAT), the proposed algorithm demonstrated improved performance by effectively extracting relevant features that capture the relationships between different sub-networks compared to other techniques examined in [32]. Motivated by the recent success of DRL in DCA, our paper takes on a challenge that has not yet been effectively addressed. Specifically, here we focus on developing a DRL framework for distributed DCA among large-scale cognitive interference networks. This framework accommodates channel reuse, continuous interference over the network graph, all without assuming complete orthogonality due to ICI.

Note that in many practical wireless communication systems, perfect orthogonality of frequency channels is difficult to achieve due to spectral leakage and hardware limitations. For instance, Wi-Fi networks operating in the 2.4 GHz band often experience overlapping channels, which leads to ICI. This phenomenon is well-documented in the literature, such as in [46], where the impact of partially overlapping channels on network performance is analyzed. Moreover, using non-orthogonal channels can lead to more efficient use of the

available spectrum. By considering overlapping channels, our model and solution are applicable to a wider range of scenarios, including those with strict spectrum reuse requirements. This approach ensures that our proposed channel allocation algorithm is robust and versatile. Finally, recent studies have continued to explore the benefits and challenges of overlapping channels. For example, [47] investigated the use of partially overlapping channels in multi-radio multi-channel wireless mesh networks, demonstrating enhanced system throughput with effective interference management. Similarly, [48] addressed joint interference cancellation and channel estimation in systems with overlapping channels, showcasing the practical applications of such models in advanced wireless networks.

Finally, the problem formulation and system model considered in this paper is applicable to a variety of practical wireless networks. These include cognitive radio networks (CRNs), where adaptive spectrum management is crucial; 6G in-X sub-networks; vehicular ad-hoc networks (VANETs), which require real-time channel allocation due to high mobility; wireless sensor networks (WSNs), operating in dynamic environments; internet of things (IoT) networks, with dense device deployments; and wireless mesh networks, which benefit from decentralized channel allocation. Our approach ensures efficient spectrum use and reliable communication across these diverse applications.

B. Main Results

Multi-agent DRL for DCA has garnered significant attention in recent years, spurred by our earlier work [22], [23]. In contrast to related recent studies that predominantly centered around relatively small-scale networks, often addressing singular sensor-controller low-power links within enclosed environments (e.g., robotic systems), the present study introduces a distributed DCA algorithm tailored for large-scale network configurations. This algorithm places paramount importance on attaining elevated channel quality (CQ) while concurrently minimizing convergence time in outdoor scenarios. Additionally, in contradistinction to existing studies that assumed orthogonal channel conditions, our approach confronts the intricate complications arising from ICI across channels, where orthogonality is not preserved. It underscores the need for minimal spectrum mobility, an imperative requirement for distributed networks supporting a multitude of users. The realistic challenging scenario considered in this paper significantly increases the search space, rendering existing algorithms inefficient.

To tackle this problem, we propose a novel MARL framework for distributed DCA, named Channel Allocation RL To Overlapped Networks (CARLTON). The CARLTON framework is based on the Centralized Training with Decentralized Execution (CTDE) paradigm, utilizing the DeepMellow value-based RL algorithm [49]. DeepMellow's design mitigates the necessity for a target network in comparison to conventional DQN algorithms. To ensure robust performance in the interference-laden environment we address, CARLTON quantizes observations to generate a QoS-type measure while maximizing a global SINR measure and guaranteeing the target QoS-SINR for each network. CARLTON incorporates a novel reward framework that trades off personal rewards

with social rewards, encouraging cooperative behavior among distinct networks incapable of information exchange. Moreover, we introduce a preprocessing mechanism designed to enhance the algorithm's adaptability to diverse network sizes. Within our specific problem domain, each network receives an exclusive sensed vector, representing only a fraction of the comprehensive state information inherent in the MARL environment. Consequently, the problem discussed herein inherently falls within the domain of Partially Observable Markov Decision Process (POMDP) problems.

The proposed CARLTON algorithm has demonstrated exceptional performance and robust generalization capabilities, outperforming various alternative methods. This includes surpassing the common random agent (RA) benchmark [20], DRL-based methods [36], [37], and, notably, outperforming the state-of-the-art method for our model, the well-known Jamming Avoidance Response (JAR) algorithm [50]. CARLTON demonstrated a remarkable approximate margin of superiority of 45% over the RA benchmark and 20% over the JAR algorithm, respectively. Furthermore, it exhibits only a slight discrepancy of approximately 2.5% in performance compared to a centralized approach computed by graph coloring. This level of effectiveness positions CARLTON as an outstanding algorithm for efficiently managing DCA among distributed cognitive networks in complex interference environments.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. Description of the System

We consider a system comprising N wireless networks represented by the set $\mathcal{N} = \{\mathcal{N}^1, \dots, \mathcal{N}^N\}$. Each network (say $\mathcal{N}^n$) accommodates M^n users, with $M^n \in \{1, 2, \dots, M_{max}\}$, and M_{max} denoting the maximum number of users among all networks. A total bandwidth (TB) is equally divided into K equal-sized overlapping channels with bandwidth B denoted by the set $\mathcal{K} = \{1, 2, \dots, K\}$ such that each network can operate on a single channel at each time slot. Orthogonality is not maintained across carriers, resulting in ICI between channels. In this context, we index the channels based on their serial numbers in $\mathcal{K}$, as depicted in Figure 1.

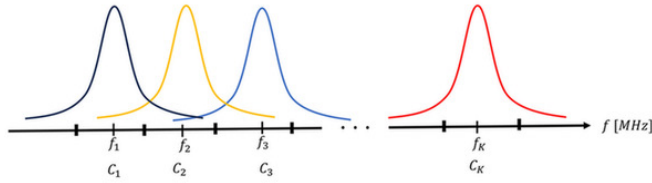


Fig. 1: An illustration of the non-orthogonal channel partition. Channel i , denoted by C_i in the figure, refers to the carrier frequency f_i .

To model the interference between networks, we employ the well-established Egli model [51]. The Egli model, developed for UHF and VHF outdoor signals, is commonly used for modeling real-world radio frequency propagation. Based on the model, the SINR of each user at each available channel is computed. The SINR accounts for the interference from the other networks in the environment, and the additive thermal

noise at the receiver. In what follows, to describe the computation, we use the indexing notations

$$(\cdot)_{m,j,k}^{l,n} \quad (1)$$

to represent a parameter regarding a pair of users, denoted as user $m \in \mathcal{N}^l$ and user $j \in \mathcal{N}^n$. The SINR is computed with respect to channel k . When the parameter pertains to a single user, irrespective of the presence of a second user, we specify only a single user. Specifically, the SINR of users within network $\mathcal{N}^n \in \mathcal{N}$ on channel $k \in \mathcal{K}$ is computed as follows. Consider users $i, j \in \mathcal{N}^n$, where user i transmits a message to user j , operating on channel $k \in \mathcal{K}$. The received SINR at the receiver of user j is given by:

$$\text{SINR}_{i,j,k}^{n,n} = \frac{\text{PR}_{i,j,k}^{n,n}}{I_T + I_{j,k}^n}, \quad (2)$$

where $\text{PR}_{i,j,k}^{n,n}$ is the signal power at the receiver of user $j \in \mathcal{N}^n$ transmitted by user $i \in \mathcal{N}^n$ on channel k , I_T is the additive thermal noise power, and $I_{j,k}^n$ is the interference power at the receiver of user $j \in \mathcal{N}^n$ on channel k . This interference is attributed to all other networks $\mathcal{N} \setminus \mathcal{N}^n$ transmitting on any possible channel, with ICI modeled by the Egli model, as computed later. Interference between users within the same network is avoided by the Medium Access Control (MAC) scheduler mechanism. The computation of the received power is given by:

$$\text{PR}_{i,j,k}^{n,n} = \text{PT}_{j,k}^n - \text{PL}_{i,j,k}^{n,n} - \chi_{i,j,k}^{n,n} - \psi_{i,j,k}^{n,n}, \quad (3)$$

where $\text{PT}_{j,k}^n$ is the transmitter power of user $j \in \mathcal{N}^n$ on channel k (possibly broadcasted signal irrespective of user $j \in \mathcal{N}^n$) in dBW, $\text{PL}_{i,j,k}^{n,n}$ is the path loss between user $j \in \mathcal{N}^n$ and user $i \in \mathcal{N}^n$ on channel k , and $\chi_{i,j,k}^{n,n}$ and $\psi_{i,j,k}^{n,n}$ stand for the small-scale fading and shadowing effects, respectively, between user $j \in \mathcal{N}^n$ and user $i \in \mathcal{N}^n$ on channel k .

The interference power from other networks $\mathcal{N} \setminus \mathcal{N}^n$ at the receiver of user $j \in \mathcal{N}^n$ on channel k is given by:

$$I_{j,k}^n = \sum_{l \in \{1, \dots, n-1, n+1, \dots, N\}} \Phi_{j,k}^{n,l} \quad (4)$$

where

$$\Phi_{j,k}^{n,l} \triangleq \sum_{m \in \{1, \dots, M^l\}} \phi_{j,m,k}^{n,l},$$

where $\phi_{j,m,k}^{n,l}$ is the interference power from user $m \in \mathcal{N}^l$, given by:

$$\phi_{j,m,k}^{n,l} = \text{PR}_{j,m,k}^{n,l} - T(k, \tilde{k}) \quad (5)$$

where $\tilde{k}$ is the channel used by network $\mathcal{N}^l$, and $T(k, \tilde{k})$ is the attenuation at the channel of interest k with respect to channel $\tilde{k}$.

B. The Objective

Let $c^n \in \{1, \dots, K\}$ represent the operating channel of network $\mathcal{N}^n$, and SINR_{j,c^n}^n be the average SINR of user $j \in \mathcal{N}^n$, computed in consideration of all messages received from

the other $M^n - 1$ users in network $\mathcal{N}^n$. Then, SINR_{j,c^n}^n is given by:

$$\text{SINR}_{j,c^n}^n = \frac{1}{M^n - 1} \sum_{\substack{i=1 \\ i \neq j}}^{M^n} \text{SINR}_{j,i,c^n}^n. \quad (6)$$

Then, the average SINR of network $\mathcal{N}^n$, denoted by $\overline{\text{SINR}}_{c^n}^n$, is given by:

$$\overline{\text{SINR}}_{c^n}^n = \frac{1}{M^n} \sum_{j=1}^{M^n} \text{SINR}_{j,c^n}^n. \quad (7)$$

The objective is to maximize the average $\overline{\text{SINR}}_{c^n}^n$ across all participating networks, subject to the constraint that the $\text{SINR}_{c^n}^n$ for each network exceeds the target QoS-SINR required for high-quality communication, defined by SINR^* :

$$\begin{aligned} \max_{\{c^n\}^N} & \frac{1}{N} \sum_{n=1}^N \overline{\text{SINR}}_{c^n}^n \\ \text{subject to } & \overline{\text{SINR}}_{c^n}^n \geq \text{SINR}^* \quad \forall n \in \mathcal{N}, \end{aligned} \quad (8)$$

Note that maximizing the average SINR with unconstrained optimization might lead to poor performance in terms of fairness across different networks. Thus, adding target QoS-SINR constraints ensures that each network meets the minimum required QoS-SINR, which is a common approach in communication networks where a minimal QoS is required when maximizing the average performance. Given the random nature of communication network channel conditions, this approach provides strong average performance for all participating networks over the long run, while still maintaining minimal QoS for each network.

In this paper, we focus on efficient communication in scenarios where users are limited to exchanging very low volumes of information within a network. Therefore, we binarize the SINR metric with respect to a target QoS-SINR sensitivity threshold SINR_T :

$$\text{BSINR}_{j,k}^n = \mathbb{1}(\text{SINR}_{j,k}^n > \text{SINR}_T), \quad (9)$$

where $\mathbb{1}$ is the indicator function, equals 1 if $\text{SINR}_{j,k}^n > \text{SINR}_T$, or equals zero otherwise.

Under this communication overhead constraint, the objective is to maximize the average BSINR metric, subject to target QoS-SINR constraints:

$$\begin{aligned} \max_{\{c^n\}^N} & \frac{1}{N} \sum_{n=1}^N \overline{\text{BSINR}}_{c^n}^n \\ \text{subject to } & \overline{\text{SINR}}_{c^n}^n \geq \text{SINR}^* \quad \forall n \in \mathcal{N}, \end{aligned} \quad (10)$$

where $\overline{\text{BSINR}}_{c^n}^n \triangleq \frac{1}{M^n} \sum_{j=1}^{M^n} \text{BSINR}_{j,c^n}^n$ is defined in a similar manner.

It should be noted that we consider users capable of measuring SINR values across all channels. While communication systems have faced challenges with implementing wideband sensing, such as increased power consumption and the need for efficient data processing, it is now becoming increasingly practical and beneficial due to technological advancements. Modern devices leverage powerful processors and sophisticated algorithms for fast and accurate signal analysis, while

software-defined radio (SDR) technology enables flexible and reconfigurable wideband sensing across various frequencies and protocols [52], [53].

Two crucial requirements for the algorithm are identified. Firstly, the algorithm must exhibit a short convergence time to swiftly adapt to the dynamic nature of the spectrum environment, ensuring timely and efficient allocation of available resources. Secondly, a distributed implementation is imperative to promote scalability, enhance flexibility, and reduce the computational burden on individual nodes, fostering a more resilient and adaptable system architecture.

III. THE CHANNEL ALLOCATION RL TO OVERLAPPED NETWORKS (CARLTON) ALGORITHM

In this section, we introduce the CARLTON algorithm designed to address the objective. We formulate the DCA problem as a MARL framework, where each network manager acts as an autonomous agent within the global framework. Consistent with standard Reinforcement Learning (RL) paradigms, the well-defined environment comprises observation, action, and reward components. Following the system model outlined in Section II, the CARLTON framework enables network managers (agents) to iteratively learn and adapt their channel allocation strategies, aiming for improved channel quality and reduced spectrum mobility until the system converges to a desirable solution. We start by providing a background on RL, and then describe CARLTON in detail.

A. Background on Reinforcement Learning

The CARLTON algorithm is trained using a value-based RL method. Specifically, value-based RL methods focus on estimating value functions, which are used to make decisions based on expected long-term rewards. One of the most common value functions is the Q-value, which represents the expected accumulated reward of taking a specific action in a given state. The Q-value is computed by:

$$Q_\pi(s, a) = E_\pi \left[\sum_{i=0}^{\infty} \gamma^i r(t+i+1) \middle| S(t) = s, A(t) = a \right].$$

In this context, π denotes the policy, r the reward, s the state, a the action, γ the discount factor, and t the time index.

A standard method for estimating $Q(s, a; \theta)$ is Q-learning, which updates the parameters after executing action $A(t) = a$ in state $S(t) = s$, receiving the immediate reward $r(t+1)$, and transitioning to state $S(t+1)$, using the following equation [54]:

$$\theta(t+1) = \theta(t) + \alpha \left(Y(t) - Q(s, a; \theta(t)) \right) \nabla_{\theta(t)} Q(s, a; \theta(t)).$$

Here, α is a constant step-size parameter, and $Y(t)$ represents the target value. The definition of $Y(t)$ may vary depending on the technique employed. In standard Q-learning, $Y(t)$ is given by:

$$Y(t) = r(t+1) + \gamma \max_a Q(S(t+1), a; \theta(t)),$$

where the current value $Q(s, a; \theta(t))$ is adjusted towards a target value $Y(t)$. In a variant of Q-learning, known as deep Q-network (DQN) [55], a multi-layered NN is utilized to map

each state s to a vector of action values $Q(s, \cdot; \theta)$, serving as the parameterized action value function. The target value $Y(t)$ is defined as follows:

$$Y(t) = r(t+1) + \gamma \max_a Q(S(t+1), a; \theta^-(t)),$$

where the parameters θ^- belong to a target network, which maintains the same architecture as the original NN with parameters θ . The parameters of the target network are copied from the original NN every τ steps, so that $\theta^- = \theta$, and remain fixed during intermediate steps. In the subsequent sections, we will provide a detailed description of all components in relation to our problem.

Note that RL was originally designed for a single agent interacting with a fully observable Markovian environment (where convergence to the optimal solution is guaranteed under certain regularity conditions). Later, RL, and particularly DRL, has been widely applied to more complex settings, including multi-agent, non-Markovian, and non-stationary environments [56], [57]. These applications have demonstrated strong performance, although convergence to the optimal solution remains an open question in general for these settings. DRL is one of the most powerful tools in AI today, particularly in multi-agent environments, across fields such as computer games, robotics, autonomous driving, and communication networks.

B. Observation Space

The observation space encompasses a fusion of BSINR measurements from all users within the network. Before the network manager makes a decision, each user transmits its individual BSINR bit for all available channels via a dedicated control channel to the network manager. This vector of BSINR values has a dimension of $\{0, 1\}^K$ (BSINR bit for each channel).

The network manager receives unique BSINR binary vectors from all users in the network. These individual vectors are then concatenated to form a BSINR matrix with dimensions $\{0, 1\}^{M^n \times K}$, where M^n corresponds to the total number of users in network $\mathcal{N}^n$. The entry $\text{BSINR}_{j,k}^n$ at row j and column k of the matrix, where $j = 1, \dots, M^n$ and $k = 1, \dots, K$, indicates whether the corresponding SINR $_{j,k}$ value is below or above the target QoS-SINR threshold, as specified in Eq. (9).

Next, the network manager performs an averaging operation on the BSINR^n matrix along the user axis to mitigate the influence of the number of users in the network. The k th entry of the vector for channel k is given by:

$$\text{QV}_k^n = \frac{1}{M^n} \sum_{j=1}^{M^n} \text{BSINR}_{j,k}^n. \quad (11)$$

This averaging process results in a vector representing the average number of users experiencing sufficient communication quality. We will refer to this vector as the Quality Vector ($\text{QV}^n \in \mathbb{R}^K$), which serves as a key observation in the simulations.

C. Action Space

The action space encompasses the set of allowable actions available to the agent at each decision point. In the current

context, the action space includes all possible frequency channels and is represented by $\mathcal{K}$. Thus, at each time step, the action of network $\mathcal{N}^n$ is defined as $a^n(t) \in \mathcal{K}$.

D. Reward

The agents are tasked with solving a combinatorial problem, namely, establishing a one-to-one mapping between each network and a channel. The objective is to reach a state where, either upon convergence (where each network remains connected to the same chosen channel until the completion of the scenario) or at the end of the scenario, the total mean quality of the selected channels across all networks is maximized, as detailed in Section II-B.

To pursue this objective, we embrace a two-part reward value, consisting of two distinct components. The first element relates to the personal reward, denoted by r_p . Its magnitude is influenced by the $\text{QV}^n(t)$ value associated with the channel quality ranking selected by the agent pre-selection cycle of all other networks at time step t . Essentially, r_p reflects a greedy mechanism aimed at selecting high-quality channels. Additionally, the personal reward experiences positive growth if the action remains unchanged. The reward computation is elucidated in Algorithm 1.

Algorithm 1 CARLTON's Personal Reward, r_p

Input: $\text{QV}^n(t)$, $a^n(t)$, $a^n(t-1)$ $\triangleright a^n(t)$ is the chosen channel at time t

Output: r_p

- 1: $v \leftarrow \text{QV}_{a^n(t)}^n(t)$ $\triangleright$ Extract the $\text{QV}^n(t)$ value of the chosen channel
- 2: **if** $v \geq \zeta$ **then**
- 3: $r_p \leftarrow r_{desired}$
- 4: **else**
- 5: $\text{SQV}^n \leftarrow \text{sort}(\text{QV}^n(t))$
- 6: $i \leftarrow 0$
- 7: **while** $i \leq K \cap v \geq \text{SQV}_i^n$ **do**
- 8: $i \leftarrow i + 1$
- $r_p \leftarrow 2(\frac{i}{K} - 0.5)$
- 9: **if** $a^n(t) == a^n(t-1)$ **then**
- 10: $r_p \leftarrow c_1 \cdot r_p$ $\triangleright c_1$ is constant > 1

The second component of the reward mechanism is designated as the *social welfare reward*, denoted by r_{sw} . This reward takes into consideration the implications of a network's actions on the individual rewards of its neighboring networks during the interval between two consecutive decision-making instances. Specifically, a network is categorized as a neighboring network if the Euclidean Distance (ED) between their network center points is less than a predetermined threshold, represented as Γ meters. In essence, r_{sw} captures the collective welfare and interdependence among the networks, accounting for the influence exerted by the behavior of each network on the rewards of others over successive iterations. The social welfare reward is calculated as the arithmetic mean of the personal rewards of all neighboring networks. The pseudo-code for the social welfare reward is outlined in Algorithm 2.

Algorithm 2 CARLTON's Social Welfare Reward, r_{sw}

Input: R_p, i, μ, Γ ▷ List of
all r_p of other networks between two consecutive decision
points, the index of the network of interest, dictionary of
all center points, and constant value

Output: r_{sw}

- 1: $n \leftarrow 0$
- 2: $r_{sw} \leftarrow 0$
- 3: **for** $j \in \{1, \dots, N\}$ **do**
- 4: **if** $\text{ED}(\mu[j], \mu[i]) \leq \Gamma \cap j \neq i$ **then**
- 5: $n \leftarrow n + 1$
- 6: $r_{sw} \leftarrow \frac{1}{n} ((n-1)r_{sw} + r_p^j)$

Finally, the total reward is a linear combination of the two reward terms:

$$r = \rho r_p + (1 - \rho) r_{sw}, \quad (12)$$

where $\rho \in [0, 1]$ is a design parameter, adjusted to solve the constrained optimization during training.

While it is straightforward to verify in simulation environments that constraints are met during testing, note that we cannot guarantee that all networks will meet QoS-SINR requirements in real-time implementations. Additionally, each network cannot independently verify whether the constraints of other networks are being met. This is because CARLTON operates in a fully distributed manner during testing, without a centralized controller or message exchanges between networks. However, this limitation is common to many distributed algorithms operating in a similar fashion, including the recently suggested distributed algorithms compared in our simulation results, and is not unique to CARLTON, nor is it specific to our constrained optimization approach. The reward design considers the QoS of neighboring networks, training the DRL model in CARLTON to incentivize networks to act cooperatively to achieve a common goal where all networks meet the QoS-SINR. A common approach to increase the likelihood of meeting constraints during testing is to train the algorithm to be more conservative than necessary, potentially sacrificing some of the objective maximization. It is possible to extend the model to allow control messages from networks that do not meet the QoS constraints, thereby helping the system achieve the common goal and ensuring that constraints are met in real-time implementations. However, this mechanism is not considered in this paper.

E. Training Procedure

CARLTON adheres to the multi-agent paradigm known as Centralized Training with Decentralized Execution (CTDE). At the beginning of each episode, each individual agent, denoted as the network manager, is assigned an exclusive serial number. This serial number determines the timing of its actions, effectively simulating a delay time characteristic of real-world scenarios. This technique is strategically employed to circumvent the potential collision of decisions among different networks and to mitigate the overall dynamic complexity of the scenario. To ensure a consistent and balanced decision-making process, the scenario horizon is explicitly defined as T times the total number of participating networks within the given scenario. Consequently, this arrangement allows each

network to possess precisely T decision points per scenario, guaranteeing equitable opportunities for each participant to influence the outcome of the scenario.

The resource allocation concerning channel frequency selection during the training phase conducted by each agent adheres to the following prescribed methodology. In particular, with a probability denoted by ϵ_b , the agents draw samples from the distribution in (13), originally introduced in the Exp3 algorithm for the non-stochastic multi-armed bandit problem [58]. Otherwise, the agents follow a greedy policy:

$$\Pr(a^n(t) = k | s^n(t)) = \frac{(1 - \alpha) e^{\beta Q(s^n(t), k; \theta)}}{\sum_{a' \in \{1, \dots, K\}} e^{\beta Q(s^n(t), a'; \theta)}} + \frac{\alpha}{K} \quad \forall k \in \mathcal{K}. \quad (13)$$

Here, $s^n(t)$ stands for the state at time step t of network $\mathcal{N}^n$, $Q(s^n(t), k; \theta)$ stands for the estimated action value by the NN with weights θ , which its architecture described below, and α and β are constants. Additionally, during the training period comprising B episodes, we annealed ϵ_b from 0.5 to 0.01 after each of the first $B/2$ episodes, and then maintained it at a steady value for the remaining half, i.e., the latter $B/2$ episodes.

Furthermore, an additional step involves the application of an action-masking technique over the $Q(s^n(t), a^n(t))$ values, which is based on the QV^n values on each channel k . For any channel in QV^n with a value of 0, the probability of selecting this channel is masked to 0, while in the event of $\text{QV}^n = \vec{0}$, no action selection is performed, and the previous action is preserved. This technique serves the purpose of preventing collisions among different networks and is mathematically executed as follows:

$$Q(s^n(t), k; \theta) = \begin{cases} -\infty, & \text{if } \text{QV}_k^n = 0, \\ Q(s^n(t), k; \theta), & \text{otherwise.} \end{cases} \quad (14)$$

Next, as is inherent in DRL value-based algorithms, our principal objective is to find an estimating function, denoted as f_θ , which approximates the action value $Q(s^n(t), a^n(t); \theta)$. This function adeptly maps any given set of state and action (s, a) to a real number within the domain of $\mathbb{R}$. The optimization of f_θ is commonly achieved through gradient descent algorithms over a differentiable loss function. In this context, we employ the Huber loss to compute our loss value. The Huber loss function is given by:

$$L_\delta(e) = \begin{cases} \frac{1}{2} e^2, & \text{if } |e| \leq \delta, \\ \delta \cdot (|e| - \frac{1}{2} \delta), & \text{otherwise,} \end{cases} \quad (15)$$

where $e^n = (r^n(t+1) + \gamma \text{mm}_\omega Q(s^n(t+1), a'; \theta) - Q(s^n(t), a^n(t); \theta))$, and mm_ω is a smooth maximum operator named Mellowmax operator with a temperature parameter ω [59]. Generally, e is the error between the target and predicted values, and δ is a discrimination hyper-parameter threshold. Subsequently, with the aid of these estimated action values, the learn policy can be derived. To enable this process, we formally define the input state as follows:

$$s^n(t) = \text{concatenate}(\text{CBR}, \text{QV}^n(t)). \quad (16)$$

Here, the term CBR represents the Channel Binary Representation, refers to the channel that the network utilizes before

executing the action. In other words, it is a binary representation of the preceding action if the action was fulfilled. The term $QV^n(t)$ stands for the QV^n at time step t .

The NN architecture of CARLTON consists of three dense layers with skip connections. The hidden layer uses the Leaky-ReLU activation function with a slope of 0.2, while the output layer employs the identity activation function. Furthermore, to initialize the weights of the NN, the Glorot-Uniform method, as introduced in the work by Glorot and Bengio [60], is applied, and the biases are initialized to zero.

The training procedure unfolds across B scenarios as follows: First, within each main training loop, we uniformly sample the number of participating networks. Subsequently, we generate a scenario following the rules outlined in Section IV-A. For each network, we then randomly assign a network number $n \in 1, \dots, N$ to define $\mathcal{N}^n$ and allocate a unique replay memory ($\mathcal{D}^n$). Next, we iterate over T time steps for each scenario, where at each step we execute actions for each network sequentially, corresponding to their assigned network number. Then, we store the relevant information in the respective unique replay buffer. Upon scenario completion, we transfer data from each $\mathcal{D}^n$ into a global replay memory (GRM). Subsequently, we uniformly sample from the GRM and train the NN by minimizing (15) N_E times.

The training procedure is illustrated in Algorithm 3, providing a comprehensive outline of the steps involved. For a clear overview of the hyperparameters utilized during the training process, please refer to Table I.

We note that during the training phase of CARLTON, networks agree on a sequential order to take actions. Practically, we achieve sequential optimization in real systems, during the inference phase, by allowing networks to update their strategies at random intervals within a given period. This randomization ensures that the probability of two networks updating simultaneously is very small. This mechanism is widely implemented in communication algorithms that require sequential updates among networks (or among users or cells, etc.), facilitating efficient and conflict-free updates across the networks (see e.g., [20], [23] and references therein).

IV. EXPERIMENTS AND DISCUSSION

In this section, we present comprehensive experimental results to illustrate the efficiency of the proposed CARLTON algorithm, as described in Section II. We commence by analyzing specific features in the training procedure of CARLTON in Subsection IV-B. Subsequently, we provide a performance comparison with state-of-the-art methods in Subsection IV-C. We first elaborate on the channel model and the system environment.

A. Illustration of the System Environment

To elaborate on the system model and its significant relevance in dynamic spectrum access systems, we present a comprehensive depiction of the system environment through practical simulation settings that inspire this research.

We first focus on the classic Egli channel model. In Section IV-C, we will demonstrate the robustness under small-scale

Algorithm 3 CARLTON's Training Procedure

Input: All hyperparameters in Table I

Input: Initialize NN weights, θ_0

Output: Trained mechanism for distributed DCA, θ^*

```

1: Create Global Replay memory (GRM) of size  $Sz$ 
2: for  $i \in \{1, \dots, B\}$  do
3:   Uniformly sample random number of networks
4:   Set a random scenario with  $N$  networks (Section IV-A)
5:   Randomly match unique network numbers
6:   Create Local Replay Memory ( $\mathcal{D}^n$ ) for each network
7:   Randomly initialize each network to some channel
8:   for  $t \in \{1, \dots, T\}$  do
9:     for  $n \in \{1, \dots, N\}$  do
10:      obtain  $QV^n(t)$ 
11:       $s^n(t) \leftarrow \text{concatenate}(\text{BCR}, QV^n(t))$ 
12:      Store  $(s^n(t-1), a^n(t-1), r^n(t), s^n(t))$  in  $\mathcal{D}^n$ 
13:      Select  $a^n(t)$  for  $\mathcal{N}^n$  based on  $f_{\theta_{i-1}}$ 
14:      Execute  $a^n(t)$  and observe  $r_p^n$ 
15:   Store all data in local  $\mathcal{D}^n$  in GRM  $\forall n \in \{1, \dots, N\}$ 
16:   for  $\{1, \dots, N_E\}$  do
17:     Uniformly sample a mini-batch of size  $bz$ ,
18:      $(s(t), a(t), r(t), s(t+1)) \sim \text{GRM}$ 
19:     Train the NN by minimizing (15) and obtain  $\theta_i$ 
```

fading and shadowing effects as well. As described in Subsection II-A, the received power at the receiver of user $j \in \mathcal{N}^n$ transmitted by user $i \in \mathcal{N}^n$ on channel k is given by

$$PR_{i,j,k}^{n,n} = PT_{j,k}^{n,n} - PL_{i,j,k}^{n,n}, \quad (17)$$

where $PT_{j,k}^{n,n}$ is the transmitter power of user $j \in \mathcal{N}^n$ on channel k (possibly broadcasted signal irrespective of user $j \in \mathcal{N}^n$) in dBW, and $PL_{i,j,k}^{n,n}$ is the path loss between user $j \in \mathcal{N}^n$ and user $i \in \mathcal{N}^n$ on channel k , which obeys the Egli model and is given by:

$$\begin{aligned}
PL_{i,j,k}^{n,n} = & 40 \log_{10}(d_{i,j}^{n,n}) \\
& - 20 \log_{10}\left(\frac{40}{f_k}\right) - 20 \log_{10}(H_{T_j}^n H_{R_i}^n) \\
& - 10 \log_{10}(G_{T_i}^n G_{R_j}^n) [dB],
\end{aligned} \quad (18)$$

where $d_{i,j}^{n,n}$ is the euclidean distance between users $j \in \mathcal{N}^n$ and $i \in \mathcal{N}^n$ in [meters], f_k is the carrier frequency of channel k in [MHz], $H_{T_j}^n$ and $H_{R_i}^n$ are the heights of the transmit and receive antennas of users $j \in \mathcal{N}^n$ and $i \in \mathcal{N}^n$ in [meters], respectively, and $G_{T_j}^n$ and $G_{R_i}^n$ are the absolute gain of the transmitter and receiver antennas of users $j \in \mathcal{N}^n$ and $i \in \mathcal{N}^n$, respectively.

The attenuation used in the simulations in this paper is given in Table II [61].

The thermal noise I_T at the receiver is calculated as follow:

$$\begin{aligned}
I_T = & k_B \cdot \text{Temp} \cdot B \cdot \text{NF} [W] = \\
& 10 \log_{10}(k_B \cdot \text{Temp} \cdot B) + 10 \log_{10}(\text{NF}) + 30 [dBm],
\end{aligned} \quad (19)$$

where k_B , Temp, B , and NF stands for Boltzmann constant, temperature, bandwidth, and noise figure. For instance, at room temperature, bandwidth of 2MHz, and $10 \log_{10}(\text{NF}) = 6$, we have $I_T = -104.9 [dBm]$.

Parameter	Value
Number of channels, K	10
Channels frequencies carriers, $\{f_1, \dots, f_k\}$ [MHz]	$\{208, \dots, 226\}$ $f_{k+1} - f_k = 2MHz\}$
Antennas gains, G_R, G_T	1
Antennas heights (H_R, H_T) [m]	1
Transmit power, PT [dBW]	2
Range of initial center point, u_1 [m]	400
Radius range, x_1, x_2 [m]	50, 500
The set of possible users per network, $\mathcal{M}$	$\{1, \dots, 15\}$
SINR threshold, $SINR^*$ [dBm]	4
Desired reward, $r_{desired}$	4
Constant, c_1	1.1
A desired threshold, ζ	0.9
Euclidean distance threshold which defines a network as a neighbor Γ [m]	500
Personal reward fraction, ρ	0.7
Distribution parameters α, β	0, 1
Huber Loss parameter, δ	1
Number of episodes during training, B	1000
MellowMax parameter, ω	0.02, if $i \leq N/2$, 0.2, otherwise
Discount factor, γ	0.9
Optimizer	Adam
Optimizer learning rate	0.00025, if $i \leq N/2$, 0.0001, otherwise
Optimizer exponential decay rate for 1st, and 2nd moment estimates, β_1, β_2	0.9, 0.999
Optimizer epsilon for numerical stability, ϵ_{Adam}	10^{-7}
Global Replay Memory size, S_z	10^5
Number of decision points per network, T (also the size of the Local replay Memory)	20
NN architecture	
Number of hidden layer	3
Nodes at each layer	128
Number of skip connection (addition after activation)	2
Hidden layer activation function	Leaky-ReLU slop of 0.2
Output activation function	Identity
Weights initialization technique	Glorot-Uniform
Bias initialization	$\vec{0}$
Number of training steps, N_E	40
Batch size, bz	32

TABLE I: A description of the hyper-parameter values used in the experiments conducted in this research.

Spectral Distance $ k - \tilde{k} $	Attenuation (dBm)
0	0
1	20
2	40
3	50
4	60
≥ 5 and $ f_k - f_{\tilde{k}} /f_k \leq 0.05$	95
else	110

TABLE II: Attenuation with respect to spectral distance, where f_k and $f_{\tilde{k}}$ represent the frequencies of channels k and $\tilde{k}$, respectively.

In each simulation instance, a specific number of networks, denoted as N , is randomly chosen. For each network, the number of users is randomly assigned, following a uniform distribution (ranging from 2 to 22 users in our simulations). The spatial distribution of users within each network is determined by a multi-variate Gaussian distribution. The expectation vector is defined by a 2D center point, and the covariance is represented by a diagonal matrix with entries $[50^2, 50^2]$ $[m^2]$. It is noteworthy that our current investigation primarily addresses a 2D spatial problem, but the methodology can be readily extended to encompass 3D dimensions. The center point for each network is determined using the procedure

outlined in Algorithm 4.

Algorithm 4 - Set Networks Center Points Locations

Input: N ▷ Number of networks
Output: μ ▷ Dictionary of all center points, $\mu \in \mathbb{R}^{2 \times N}$

- 1: $\mu \leftarrow \{\}$ ▷ Define empty dictionary
- 2: **for** $i \in \{1, \dots, N\}$ **do**
- 3: **if** $i == 0$ **then**
- 4: $\mu_x^0 \leftarrow u_x$ s.t $U_x \sim U[-u_1N, u_1N]$
- 5: $\mu_y^0 \leftarrow u_y$ s.t $U_y \sim U[-u_1N, u_1N]$
- 6: **else**
- 7: $\mu^c = \text{choose random center point from } \mu$
- 8: $r \leftarrow \text{sample from } U[x_1, x_2]$
- 9: $\theta \leftarrow \text{sample from } U[0, 2\pi]$
- 10: $\mu_x^i \leftarrow \mu_x^c + r \cos(\theta)$
- 11: $\mu_y^i \leftarrow \mu_y^c + r \sin(\theta)$
- $\mu[i] \leftarrow (\mu_x^i, \mu_y^i)$

In our simulation environment, we incorporate the concept that each network includes a specific user designated as the *network manager*, aligning with the common practice in 5G technology where centralized units oversee local networks. The network manager is selected as the user within the network whose total Euclidean distance from all other users in the same network is minimized. This selection is based on the user's ability to maintain the shortest average distance to all other users within the given network. The role of the network manager is pivotal, taking on the responsibility of gathering information from other users within the network and disseminating decisions as feedback. This feedback mechanism fosters effective communication and coordination among users, allowing the network manager to serve as a central point for control and information exchange within its respective network, using a dedicated control channel. By fulfilling this crucial function, the network manager significantly contributes to optimizing the overall efficiency of the network's operations. Figure 2 illustrates a representative scenario, featuring a possible configuration of networks, each accompanied by its designated network manager.

B. CARLTON's Training Results

CARLTON undergoes two distinct training scenarios, one involving the action masking approach, as referenced in (14), and the other without it. As expected, the incorporation of masking during action selection results in higher accumulated rewards during the initial phases of training (Episodes ≤ 400). This improvement stems from the reduction of unnecessary exploration, as illustrated in Figure 3. However, as training progresses and convergence is achieved (Episodes ≥ 500), both techniques yield comparable results. It is important to note that both approaches remain applicable when the messages exchanged between network users and the network manager are transmitted through a control channel, which is assigned and freed for communication at any given time.

On the other hand, in situations where no dedicated control channel is available, the utilization of masking becomes essential to ensure the existence of a functional communication link between the users and the network manager. Thus, it

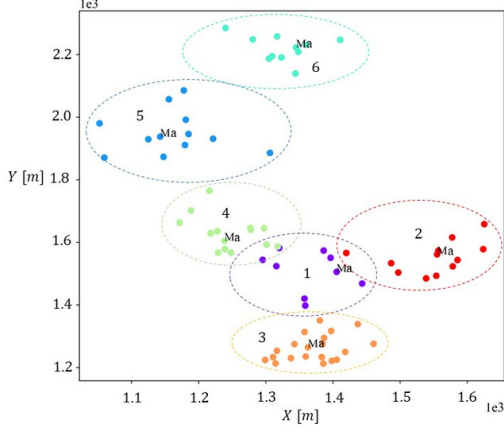


Fig. 2: An illustration of the simulation environment involving six networks. Each color corresponds to a distinct network, distinguished by a unique serial number. The symbol 'Ma' denotes the network manager present at each respective network.

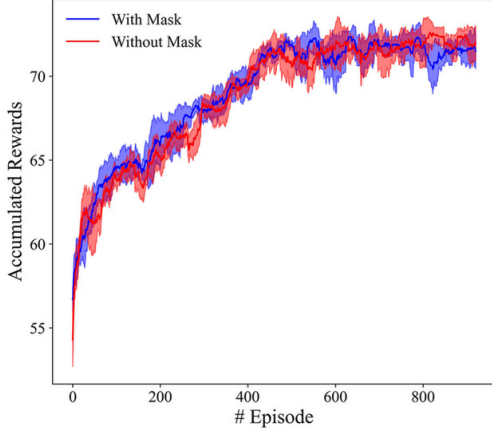


Fig. 3: CARLTON performance: The accumulated rewards as function of episode number during training for implementation of with and without masking approach. In a case of perfect game the maximum value is 88.

guarantees that there is at least some form of connection facilitating communication and data exchange. In addition, using the masking approach not only depends on the presence or absence of a control channel for communication but also helps to fast convergence in case of higher action space, as shown in previous work [62]–[65]. When directing our attention to the physical results obtained during the training process, it is evident that CARLTON exhibits noteworthy enhancements in the mean, median, and minimum values of the channel quality. The improvement of these values, as defined in (20), are shown in Figure 4. Furthermore, upon convergence, it is conspicuous that the agents have acquired the ability to cooperate effectively, resulting in a scenario where the lowest channel quality (min_{CQ}) of some agents in the system remains above 0.95 percent. Here, $c^n(t)$ is the channel of network n at time step t .

$$\begin{aligned} CQ &= \left[QV_{c^1(T)}^1(T), \dots, QV_{c^N(T)}^N(T) \right], \\ \overline{CQ} &= \mathbb{E}[CQ], \\ \widetilde{CQ} &= \text{Median}(CQ), \\ min_{CQ} &= \min(CQ). \end{aligned} \quad (20)$$

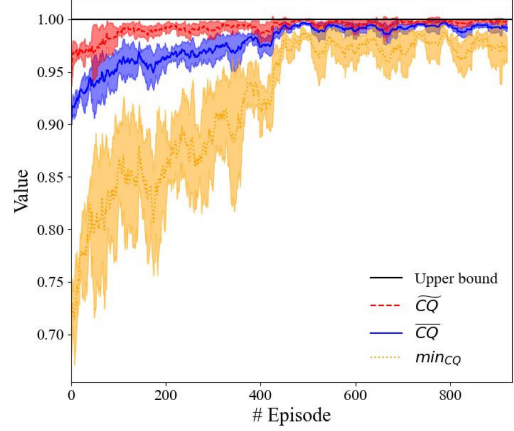


Fig. 4: CARLTON performance: The empirical results of the mean ($\overline{CQ}$), median ($\widetilde{CQ}$), minimum (min_{CQ}) values of channel quality during training.

Similarly, we have investigated the progress observed during training regarding supplementary physical parameters. The first additional parameter is the average number of channel changes (ANCC), defined as the average number of channel changes until convergence per network. The ANCC score (ANCCS) is given by:

$$ANCCS = 1 - \frac{ANCC}{T}. \quad (21)$$

The second parameter is the convergence time (CT), which refers to the time of the last channel change made by one of the networks during the scenario. The convergence time score (CTS) is calculated as follows:

$$CTS = 1 - \frac{CT}{T \cdot N}. \quad (22)$$

The third parameter, denoted as spectrum efficiency (SE), is intended to assess the spectral distortion resulting from the utilization of networks after convergence or at the end of the episode. A higher value of SE parameter indicates that the integration of a new network into the group becomes an easier task by providing more channels with high quality to operate on, along with causing minimal distortion to the convergences solution achieved by the earlier participants. The SE score (SES) is computed as follows:

$$\text{SES} = \mathbb{E}[\Psi] = \frac{1}{N} \sum_{n=1}^N \Psi^n,$$

where

$$\Psi^n = \left(\sum_{k=1}^K (\text{QV}_k^n(T))^2 \right)^{0.5} / K^{0.5}. \quad (23)$$

The last parameter that we define is a linear weighted score (WS) of the parameters of interest, given by:

$$\text{WS} = w_1 \cdot \overline{\text{CQ}} + w_2 \cdot \text{ANCCS} + w_3 \cdot \text{CTS} + w_4 \cdot \text{SES}, \quad (24)$$

where we set w_1, w_2, w_3, w_4 in the simulations to 0.4, 0.1, 0.4, 0.1, respectively.

A higher weight ($w_1 = 0.4$) is assigned to $\overline{\text{CQ}}$ because maintaining high channel quality is critical for ensuring reliable communication and minimizing interference, directly impacting the network's ability to meet QoS requirements. Similarly, a high weight ($w_3 = 0.4$) is given to CTS, as faster convergence times are crucial for adapting to dynamic network conditions and ensuring timely channel allocations, which helps in maintaining efficient network operations, especially in environments with high user mobility. A lower weight ($w_2 = 0.1$) is assigned to ANCCS because while reducing the number of channel changes is beneficial for stability and reducing control overhead, it is less critical compared to ensuring high channel quality and fast convergence. The SES also has a lower weight ($w_4 = 0.1$), reflecting its importance in optimizing spectrum usage without significantly compromising other critical performance metrics. While high spectrum efficiency is desirable, it should not outweigh the need for good channel quality and quick convergence.

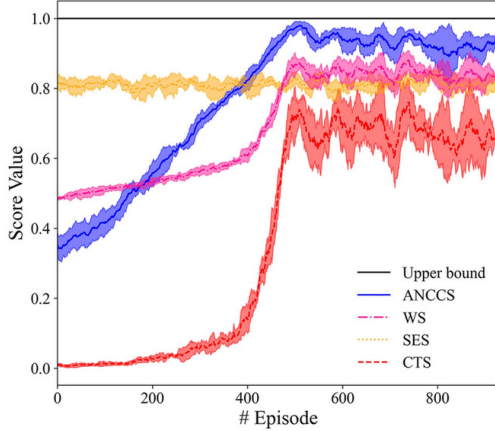


Fig. 5: CARLTON performance: The score values of the average number of channel changes (ANCC), convergence time (CT), spectrum efficiency (SE), and the weighted score (WS) during training.

As apparent from Figure 5, all parameter scores, except SE, demonstrate improvement. These results align with our reward function, as specified in (12), wherein the impact of the CT and ANCC on the reward value is given in line 10 in Algorithm 1. Moreover, in view of the absence of a direct motivation to enhance the SE score, the algorithm possesses the freedom

to establish an SE score that harmonizes with the converged policy, and results in an average value of approximately 0.8 for the SE score.

Moreover, we conducted an analysis of the influence of the personal reward weight, denoted as ρ in (12), on the algorithm's physical behavior. All options were tested over 30 games for each scenario, where the scenarios differed by the number of networks participating, ranging from 2 to 15 networks. In total, this amounted to 420 games. Despite the training process being limited to scenarios involving a maximum of 7 networks, our examination extends to include the evaluation of the algorithm's generalization capabilities across scenarios with a larger number of participating networks. It was observed that, with respect to the metric WS as depicted in Figure 6, higher values demonstrated superior performance and enhanced generality. Conversely, in the context of CQ and $\min_{CQ}$, lower values exhibited better performance, as evidenced by the expectation of the mean value displayed in Figure 7. This disparity in results can be attributed to the fact that higher values lead to more aggressive policies, resulting in quicker convergence. On the other hand, lower values facilitated increased cooperation and fairness among the participating networks, leading to a slower but more stable mechanism. A value of 0.7 was found to effectively strike a balance between these two contrasting aspects, rendering it an effective selection. In the context of cooperative missions, this equilibrium point is particularly noteworthy, as it arises from the operational constraints of decentralized agents without information exchange capabilities, effectively achieving equilibrium between self-interest and social responsibility.

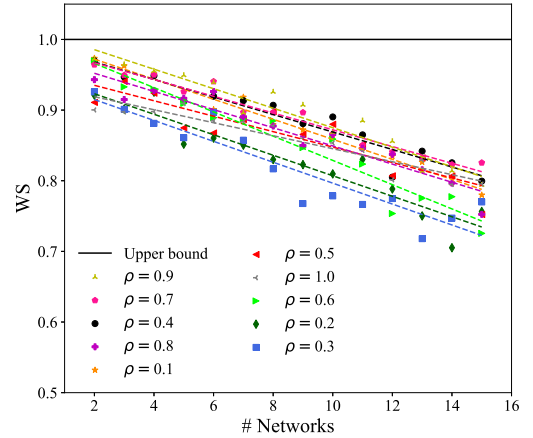


Fig. 6: CARLTON performance: The weighted score value, WS, as a function of the number of networks participating in the scenario for different values of ρ .

Afterward, we conducted an analysis pertaining to the influence of a designated threshold parameter denoted as Γ , which serves to establish the criteria for considering networks as neighbors. To briefly recapitulate, if the ED between the central points of two networks falls below the threshold Γ , both networks are deemed neighbors. These neighbor networks influence each other's rewards through the parameter denoted as r_{sw} , as shown in (12) and elaborated in Algorithm 2. In the scenario where Γ approaches zero ($\Gamma \rightarrow 0 [m]$), each network progressively exhibits more self-centered behavior, prioritizing

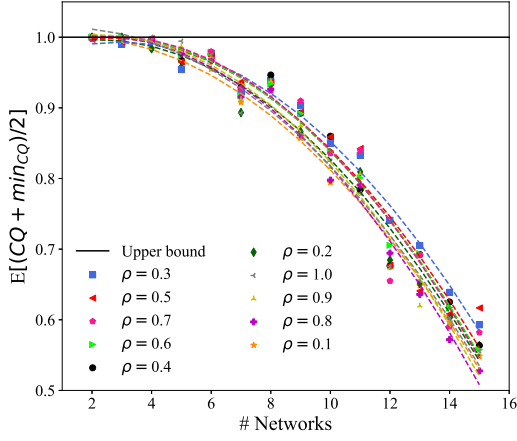


Fig. 7: CARLTON performance: the expectation of the mean value of CQ and $\min_{CQ}$ as a function of the number of networks participating in the scenario for different values of ρ .

its individual performance exclusively. Conversely, when Γ tends towards infinity ($\Gamma \rightarrow \infty$), each network gradually adopts a more socialistic stance, with its reward encompassing a broader spectrum of network performances, including those with which it shares no direct impact on their QVs. For example, when the Euclidean Distance (ED) between the central points equal to 700[m], there exists a 3% probability that the two networks will exert an influence on each other's QVs. Our analysis unveiled that, concerning WS performance, designating a network as a neighbor when the distance falls below 500 meters (corresponding to a probability of 50% that both networks influence each other's QVs) within the environmental framework described in Section IV-A), yielded the highest average performance across all tested scenarios, as depicted in Figure 8.

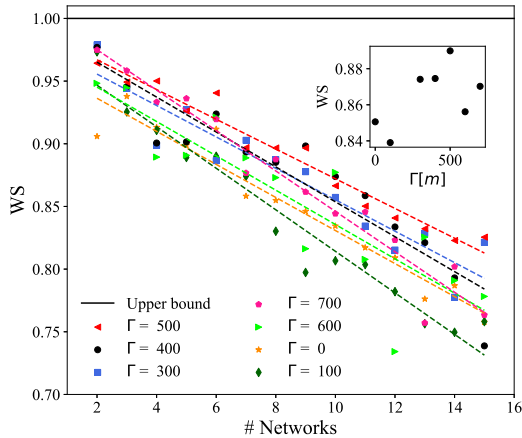


Fig. 8: CARLTON performance: The weighted score value, WS, as a function of the number of networks participating in the scenario for different values function of Γ [m]. Inset: The average value of WS across all scenarios as function of Γ [m].

On the contrary, when we shift our attention to the expectation value of $[(CQ + \min_{CQ})/2]$, it becomes evident that $\Gamma = 400$ meters (representing a probability of 86% that the networks impact each other's QVs) yields the most favorable average value across the entire domain under examination, as

illustrated in Figure 9. These findings underscore the notion that, in order to achieve a high level of social performance encompassing all participants within the given scenario, a network should only consider interactions with those networks that have a high probability of interaction. This concept not only benefits individual performance but also augments the performance of the entire system. Conversely, it is evident that adopting and learning a selfish policy ($\Gamma = 0$) exhibits the least favorable performance across all scenarios examined. It is noteworthy that when taking into account the average of both parameters, $\Gamma = 500$ meters stands out as the leading choice.

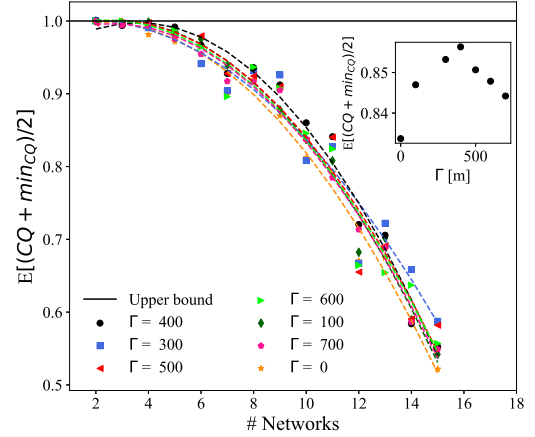


Fig. 9: CARLTON performance: The expectation of the mean value of CQ and $\min_{CQ}$ as a function of the number of networks participating in the scenario for different values of Γ [m]. Inset: The average value of $E[(CQ + \min_{CQ})/2]$ across all scenarios as function of Γ [m].

Subsequently, we sought to further enhance the performance of our proposed algorithm by implementing a post-processing step in the decision-making mechanism. Given the algorithm's generalization across diverse and extensive domains involving varying numbers of participating networks, certain instances of unnecessary channel switches were observed. For example, in a 2-network scenario, the algorithm would switch from a channel with a CQ of 1.0 to another channel with the same CQ value, even when this switching had no impact on the other participant. Such occurrences are a result of the algorithm's generalization mechanism, which has learned to create sparsity and adopt policies that aim to provide favorable performance in numerous situations. To address this issue, we introduced a post-processing step as follows: If the CQ of the desired channel is not greater than the CQ of the current channel by at least $\varphi\%$, the algorithm maintains its current channel, and effectively abstained from unnecessary changes. However, if the difference in CQ values exceeds the specified threshold, the algorithm follows the proposed action and executes the channel switch accordingly. The baseline algorithm, without post-processing, is denoted by $\varphi = \text{None}$. Furthermore, in multi-agent systems with a high number of members, it can be challenging to account for every possible scenario. Consequently, the agents tend to learn policies that exhibit bias toward the scenarios within the training set. Applying a threshold based on channel quality can reduce unnecessary spectrum mobility, which serves as a post-processing block.

To illustrate, consider a scenario in which a single network, initially operating on a randomly selected channel, might transition to an alternative channel without any rational reason, a behavior likely stemming from the biased policy learned by the agents.

The findings illustrate a correlation between the increase in the weighted score as the threshold value, φ , increases, as evident from the observations in Figure 10. Conversely, there is a decrease in the channel quality, as represented in Figure 11. This indicates that the inclusion of the threshold significantly expedites the convergence time, albeit at the cost of achieving a better solution concerning channel quality. The selection of the optimal threshold value is contingent upon two crucial factors, the size of the network, where the switch to another channel can be expensive in terms of communication operations, and the quality threshold set by the user. These considerations emphasize the significance of setting a thoughtful threshold to achieve a trade-off between convergence speed and channel quality. The introduction of a post-decision mechanism, as an optional modification, holds promise for overall performance enhancement, thereby warranting further investigation and experimentation to assess its potential benefits fully.

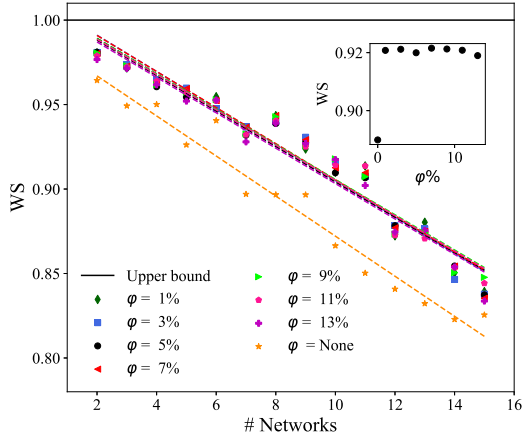


Fig. 10: CARLTON performance: The expectation of the weighted score, WS, as a function of participating network at each scenario. Inset: The average value of WS across all scenarios as function of the threshold value φ , 0 stands for None.

C. Performance Comparison

We now present a comprehensive comparative analysis to assess the performance of CARLTON compared to other well-known algorithms. To establish an upper bound, we considered an algorithm based on graph coloring, which operates in a centralized manner and is not suitable for distributed implementations. We extend the traditional graph vertex coloring method to account for overlapping channels, where interference occurs not only between the same channels but also between multiple neighboring channels. The interference decreases as the frequency interval between channels increases. To model this, we set a threshold for acceptable interference between users, which is adjustable during the search. Channels are then grouped into sets, where each set can be considered a color. When assigning channels (colors)

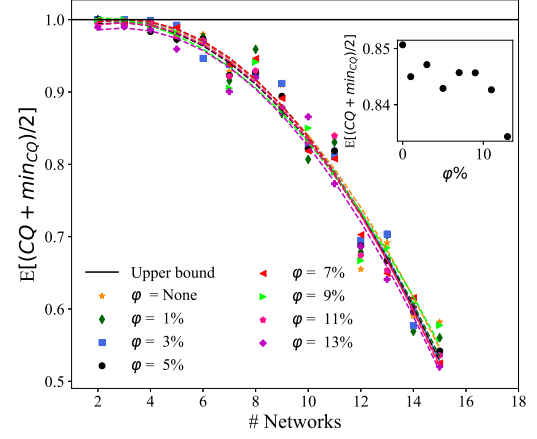


Fig. 11: CARLTON performance: The expectation of the average between CQ and $\min_{CQ}$ as a function of participating network at each scenario. Inset: The average value of $E[(CQ + \min_{CQ})/2]$ across all scenarios as function of the threshold value φ , 0 stands for None.

to the vertices (networks) of the graph, we ensure that the interference between any two adjacent vertices does not exceed the predetermined threshold. Since this optimization problem is intractable, we implemented a heuristic centralized iterative local search to find the best solution that maximizes the objective, which is a common approach in the literature [66]. Our method involves iteratively improving an initial solution by exploring its local neighborhood, aiming to minimize interference while maximizing channel utilization. Furthermore, we conducted a comprehensive evaluation of CARLTON against the following alternative approaches: (i) The Jammer Avoidance Response (JAR) [50], which is a bio-inspired algorithm, originally derived from a behavior observed in certain species of weakly electric fish. These fish generate electric fields for navigation, communication, and prey detection. When two fish with similar electric organ discharge (EOD) frequencies come into close proximity, they can interfere with each other's electric signals, a phenomenon known as jamming. To avoid this jamming, the fish adjust their EOD frequencies: if two fish detect that their EOD frequencies are too similar, one will shift its frequency up and the other will shift its frequency down to avoid interference. The main principles of the JAR algorithm in communication networks include the ability to sense interference between EODs from multiple sources, the capacity to change EOD frequencies within a specific interval to minimize this interference, and the uniform application of the JAR across all participating individuals within the spectrum. The key characteristics of JAR are the spectral range, which defines the frequencies over which the JAR operates, and the limits of spectral change, specifying the minimum and maximum adjustments in frequency that can be made to avoid jamming. In our study, we used a spectral range of 208-226 MHz, with fixed adjustments of ± 2 MHz, and an initial frequency randomly chosen from the set of channel frequency carriers. To achieve faster convergence, we required an enhancement of at least 0.05 in CQ before performing frequency adjustment. This approach has demonstrated state-of-the-art results within the considered model compared to existing methods. (ii) The second algorithm we compare to is the DRL-based multi-agent double deep Q-network (DDQN) for DCA

(MADDQN-DCA), introduced in [36]. In MADDQN-DCA, the algorithm involves training agents in a simulated environment to make decisions about channel selection, aiming to enhance the overall performance in dense deployments of 6G in-X subnetworks. A standard DDQN value-based technique was employed using a centralized training with decentralized execution approach to determine the optimal parameters of a DCA policy, represented as a fully connected NN. (iii) The third algorithm is the intelligent DRL-based spectrum access (IDSA), introduced in [37]. In IDSA, a dueling-DQN value-based technique was employed to optimize a DCA policy for imperfect feedback in multiuser wireless networks. This approach was designed as a distributed DRL system with the goal of enhancing the network performance. The algorithm addresses partial observability and imperfect feedback for each agent by incorporating a Gated Recurrent Unit (GRU) into the network architecture and utilizing a unique replay buffer for incomplete experiences. The training technique used in IDSA is independent Q-learning (IQL). (iv) Additionally, we evaluated against the well-known Random Agent (RA) [20] static initialization, where we randomly set the initial channel and refrain from making any further channel switches. The comparison with these alternative algorithms provided valuable insights into the effectiveness and advantages of CARLTON within the considered model.

In a manner analogous to the comparison involving ρ , all algorithms underwent testing across the identical set of 420 games. The obtained results demonstrated the superior performance of CARLTON in terms of both the WS metric and the expectation of $[CQ + \min_{CQ}]/2$ values, as illustrated in Figure 12 and Figure 13, respectively.

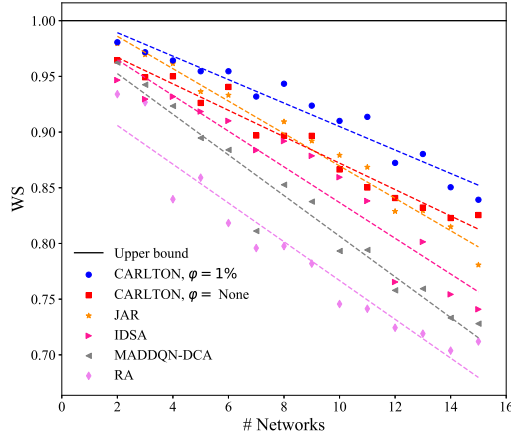


Fig. 12: Algorithm comparison: The weighted score value, WS, as a function of the number of networks participating in the scenario.

Nevertheless, as shown in Figure 14, the convergence time score demonstrates higher values in the case of CARLTON with $\varphi = 1\%$. Furthermore, CARLTON displayed a marginal difference of only 1% compared to the centralized graph-coloring approach in the in-sample domain (i.e., $\#Networks < 7$), underscoring its efficacy in distributed systems, as illustrated in Figure 15.

The superior performance of CARLTON over the other algorithms can be explained as follows: The RA algorithm

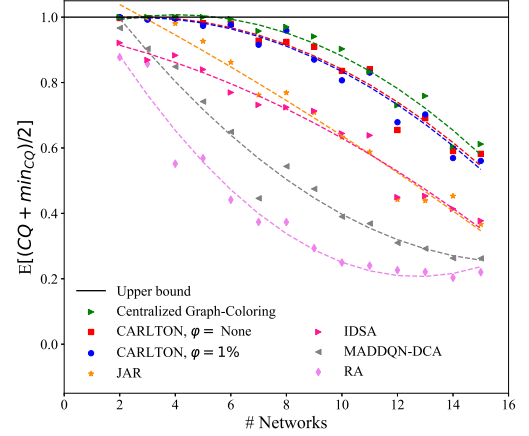


Fig. 13: Algorithm comparison: The expectation of the mean value of CQ and $\min_{CQ}$ as a function of the number of networks participating in the scenario.

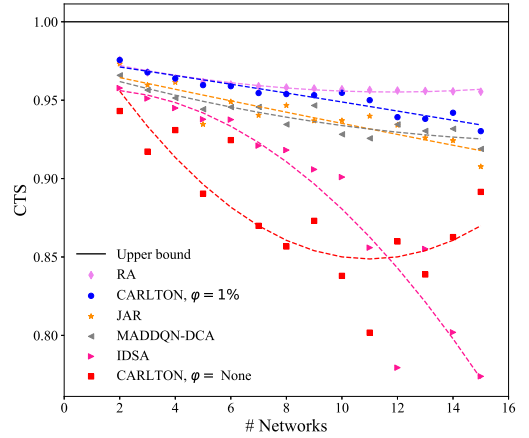


Fig. 14: Algorithm comparison: The convergence time score (CTS) as a function of the number of networks participating in the scenario.

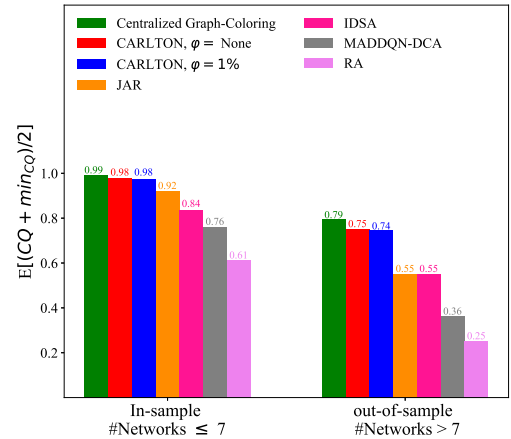


Fig. 15: Algorithm comparison: The average value of $E[(CQ + \min_{CQ})/2]$ across all scenarios for each algorithm.

exhibits a high CTS but suffers from poor channel quality. This characteristic behavior positions it as the poorest algorithm, as it does not consider channel quality. The JAR algorithm

attempts to maximize the channel quality of each participant. However, it still behaves in a self-centered manner, with each participant following an egocentric policy. This approach is suboptimal when dependencies exist between participants. Additionally, the JAR algorithm heavily relies on the initial channel of each participant. It can only modify the operating channel by moving left or right, without the ability to swap channels with other participants. This limitation hampers performance as the number of participants increases. However, relaxing this limitation can cause the JAR algorithm to exhibit unconverged behavior within the defined time interval, set here as 20 steps per participant. On the contrary, our proposed CARLTON algorithm operates without such constraints, employing a more socially aware policy that considers interdependencies among participants. Given this wider consideration, it still sustains to achieves convergence within the defined time interval. CARLTON also outperforms the DRL-based MADDQN-DCA and IDSA algorithms. According to [36], the proposed solution shows diminished performance when inter-agent information flow is not possible. Although the method outperforms a random agent as reported in [36] and in our evaluation, it performs worse in terms of WS and CQ. The use of DDQN with a very shallow NN introduces significant challenges in convergence, particularly in the context of non-orthogonal channels. In IDSA, the method employs a more sophisticated NN architecture and advanced DQN techniques compared to MADDQN-DCA, which may explain its superior performance relative to MADDQN-DCA. However, it uses Independent Q-Learning (IQL) during policy training, which is known to produce sub-optimal solutions and often faces convergence difficulties when handling large state spaces, particularly in the complex model considered here.

Next, we repeated these simulations under small-scale fading and shadowing effects as described in (3), characterized by Rice and log-normal distributions, respectively. Specifically, the Rice distribution parameter was set to 4, and the mean of the log-normal random variable's natural logarithm was set to 0. The results showed slight degradation across all algorithms, as expected, but once again, CARLTON demonstrated superior performance. Specifically, its efficiency in terms of the WS metric and the expectation of $[CQ + \min_{CQ}]/2$ values is illustrated in Figure 16 and Figure 17. In Figure 18, the convergence time score is higher for CARLTON with $\varphi = 1\%$. Additionally, CARLTON nearly matches the performance of the centralized graph-coloring approach, as shown in Figure 19.

Finally, the obtained results unequivocally demonstrate the exceptional performance of CARLTON, even in scenarios where it was not explicitly trained on (i.e., $\#Networks > 7$, named out-of-sample), thereby highlighting its remarkable generality. This remarkable adaptability can be attributed to the underlying concept that scenarios with a high number of networks in certain constellations can be effectively represented by combinations of multiple scenarios involving a lower number of networks. Consequently, the agent exhibits cooperation not with all networks but rather selectively with a subset, likely comprising fewer than 7 networks, which are considered its neighboring networks. This premise elucidates the compelling rationale behind the observed excellent generality.

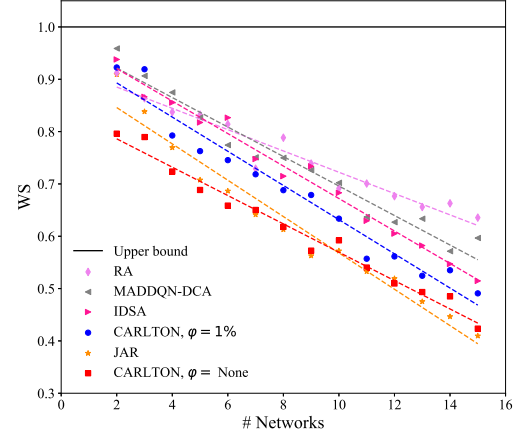


Fig. 16: Algorithm comparison under small-scale fading and shadowing effects: The weighted score value, WS, as a function of the number of networks participating in the scenario.

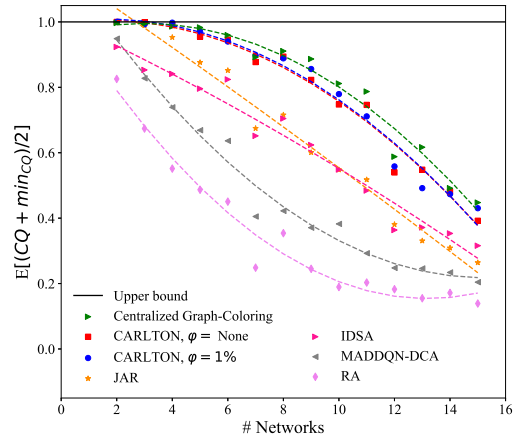


Fig. 17: Algorithm comparison under small-scale fading and shadowing effects: The expectation of the mean value of CQ and $\min_{CQ}$ as a function of the number of networks participating in the scenario.

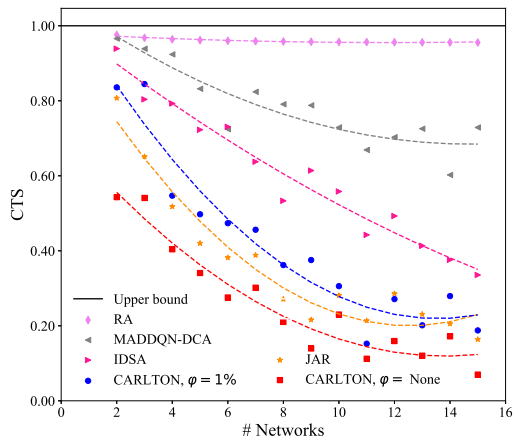


Fig. 18: Algorithm comparison under small-scale fading and shadowing effects: The convergence time score (CTS) as a function of the number of networks participating in the scenario.

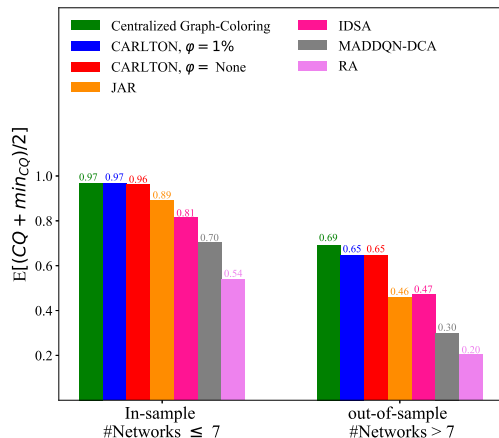


Fig. 19: Algorithm comparison under small-scale fading and shadowing effects: The average value of $E[(CQ + \min_{CQ})/2]$ across all scenarios for each algorithm.

V. CONCLUSION

We have introduced a novel distributed DCA algorithm, dubbed CARLTON, designed for multi-agent systems within the framework of MARL. CARLTON is specifically tailored for scenarios involving non-orthogonal operating channels with ICI, necessitating cooperation among participating networks to enhance spectral efficiency. The key emphasis of CARLTON lies in networks with a substantial number of users, aiming to achieve both high channel quality and rapid convergence. Our experimental results highlight the superior performance of CARLTON compared to alternative approaches, exhibiting only a marginal trade-off in channel quality when compared to the centralized-based graph-coloring approach. Moreover, CARLTON demonstrates remarkable generalization capabilities, effectively handling scenarios that were not encountered during training (out-of-sample scenarios). This effectiveness positions CARLTON as an exceptional algorithm for proficiently managing DCA among distributed cognitive networks in complex interference environments.

REFERENCES

- [1] K. Liu and Q. Zhao, "Cooperative game in dynamic spectrum access with unknown model and imperfect sensing," *IEEE Transactions on Wireless Communications*, vol. 11, no. 4, pp. 1596–1604, 2012.
- [2] C. Tekin and M. Liu, "Online learning of rested and restless bandits," *IEEE Transactions on Information Theory*, vol. 58, no. 8, pp. 5588–5611, 2012.
- [3] H. Liu, K. Liu, and Q. Zhao, "Learning in a changing world: Restless multiarmed bandit with unknown dynamics," *IEEE Transactions on Information Theory*, vol. 59, no. 3, pp. 1902–1916, 2013.
- [4] N. Nayar, D. Kalathil, and R. Jain, "On regret-optimal learning in decentralized multiplayer multiarmed bandits," *IEEE Transactions on Control of Network Systems*, vol. 5, no. 1, pp. 597–606, 2016.
- [5] O. Avner and S. Mannor, "Multi-user lax communications: a multi-armed bandit approach," in *IEEE INFOCOM 2016-The 35th Annual IEEE International Conference on Computer Communications*, pp. 1–9, IEEE, 2016.
- [6] I. Bistriz and A. Leshem, "Distributed multi-player bandits—a game of thrones approach," in *Advances in Neural Information Processing Systems*, pp. 7222–7232, 2018.
- [7] T. Gafni and K. Cohen, "Learning in restless multi-armed bandits via adaptive arm sequencing rules," *IEEE Transactions on Automatic Control*, 2020.
- [8] W. Wang, A. Leshem, D. Niyato, and Z. Han, "Decentralized learning for channel allocation in iot networks over unlicensed bandwidth as a contextual multi-player multi-armed bandit game," *IEEE Transactions on Wireless Communications*, vol. 21, no. 5, pp. 3162–3178, 2021.
- [9] T. Gafni, M. Yemini, and K. Cohen, "Learning in restless bandits under exogenous global markov process," *IEEE Transactions on Signal Processing*, vol. 70, pp. 5679–5693, 2022.
- [10] A. Leshem, E. Zehavi, and Y. Yaffe, "Multichannel opportunistic carrier sensing for stable channel access control in cognitive radio systems," *IEEE Journal on Selected Areas in Communications*, vol. 30, no. 1, pp. 82–95, 2012.
- [11] T. Gafni and K. Cohen, "Distributed learning over markovian fading channels for stable spectrum access," *IEEE Access*, vol. 10, pp. 46652–46669, 2022.
- [12] Z. Han, Z. Ji, and K. R. Liu, "Fair multiuser channel allocation for OFDMA networks using Nash bargaining solutions and coalitions," *IEEE Transactions on Communications*, vol. 53, no. 8, pp. 1366–1376, 2005.
- [13] I. Menache and N. Shimkin, "Rate-based equilibria in collision channels with fading," *IEEE Journal on Selected Areas in Communications*, vol. 26, no. 7, pp. 1070–1077, 2008.
- [14] I. Menache and A. Ozdaglar, "Network games: Theory, models, and dynamics," *Synthesis Lectures on Communication Networks*, vol. 4, no. 1, pp. 1–159, 2011.
- [15] L. M. Law, J. Huang, and M. Liu, "Price of anarchy for congestion games in cognitive radio networks," *IEEE Transactions on Wireless Communications*, vol. 11, no. 10, pp. 3778–3787, 2012.
- [16] K. Cohen, A. Leshem, and E. Zehavi, "Game theoretic aspects of the multi-channel ALOHA protocol in cognitive radio networks," *IEEE Journal on Selected Areas in Communications*, vol. 31, pp. 2276–2288, 2013.
- [17] H. Wu, C. Zhu, R. J. La, X. Liu, and Y. Zhang, "Fasa: Accelerated S-ALOHA using access history for event-driven M2M communications," *IEEE/ACM Transactions on Networking (TON)*, vol. 21, no. 6, pp. 1904–1917, 2013.
- [18] C. Singh, A. Kumar, and R. Sundaresan, "Combined base station association and power control in multichannel cellular networks," *IEEE/ACM Transactions on Networking*, vol. 24, no. 2, pp. 1065–1080, 2016.
- [19] K. Cohen and A. Leshem, "Distributed game-theoretic optimization and management of multichannel aloha networks," *IEEE/ACM Transactions on Networking*, vol. 24, no. 3, pp. 1718–1731, 2016.
- [20] K. Cohen, A. Nedić, and R. Srikant, "Distributed learning algorithms for spectrum sharing in spatial random access wireless networks," *IEEE Transactions on Automatic Control*, vol. 62, no. 6, pp. 2854–2869, 2017.
- [21] D. Malachi and K. Cohen, "Queue and channel-based aloha algorithm in multichannel wireless networks," *IEEE Wireless Communications Letters*, vol. 9, no. 8, pp. 1309–1313, 2020.
- [22] O. Naparstek and K. Cohen, "Deep multi-user reinforcement learning for dynamic spectrum access in multichannel wireless networks," in *GLOBECOM 2017-2017 IEEE Global Communications Conference*, pp. 1–7, IEEE, 2017.
- [23] O. Naparstek and K. Cohen, "Deep multi-user reinforcement learning for distributed dynamic spectrum access," *IEEE transactions on wireless communications*, vol. 18, no. 1, pp. 310–323, 2018.
- [24] H.-H. Chang, H. Song, Y. Yi, J. Zhang, H. He, and L. Liu, "Distributive dynamic spectrum access through deep reinforcement learning: A reservoir computing-based approach," *IEEE Internet of Things Journal*, vol. 6, no. 2, pp. 1938–1948, 2018.
- [25] P. Yang, L. Li, J. Yin, H. Zhang, W. Liang, W. Chen, and Z. Han, "Dynamic spectrum access in cognitive radio networks using deep reinforcement learning and evolutionary game," in *2018 IEEE/CIC International Conference on Communications in China (ICCC)*, pp. 405–409, IEEE, 2018.
- [26] W. Su, J. Lin, K. Chen, L. Xiao, and C. En, "Reinforcement learning-based adaptive modulation and coding for efficient underwater communications," *IEEE access*, vol. 7, pp. 67539–67550, 2019.
- [27] Y. Zhang, X. Wang, and Y. Xu, "Energy-efficient resource allocation in uplink noma systems with deep reinforcement learning," in *2019 11th international conference on wireless communications and signal processing (WCSP)*, pp. 1–6, IEEE, 2019.
- [28] D. Lee, Y. G. Sun, S. H. Kim, I. Sim, Y. M. Hwang, Y. Shin, D. I. Kim, and J. Y. Kim, "Dqn-based adaptive modulation scheme over wireless communication channels," *IEEE Communications Letters*, vol. 24, no. 6, pp. 1289–1293, 2020.
- [29] W. Ning, X. Huang, K. Yang, F. Wu, and S. Leng, "Reinforcement learning enabled cooperative spectrum sensing in cognitive radio networks," *Journal of Communications and Networks*, vol. 22, no. 1, pp. 12–22, 2020.
- [30] J. Xu, H. Lou, W. Zhang, and G. Sang, "An intelligent anti-jamming scheme for cognitive radio based on deep reinforcement learning," *IEEE Access*, vol. 8, pp. 202563–202572, 2020.

- [31] F. Obite, A. D. Usman, and E. Okafor, "An overview of deep reinforcement learning for spectrum sensing in cognitive radio networks," *Digital Signal Processing*, vol. 113, p. 103014, 2021.
- [32] X. Du, T. Wang, Q. Feng, C. Ye, T. Tao, L. Wang, Y. Shi, and M. Chen, "Multi-agent reinforcement learning for dynamic resource management in 6g in-x subnetworks," *IEEE Transactions on Wireless Communications*, vol. 22, no. 3, pp. 1900–1914, 2022.
- [33] R. Adeogun and G. Berardinelli, "Multi-agent dynamic resource allocation in 6g in-x subnetworks with limited sensing information," *Sensors*, vol. 22, no. 13, p. 5062, 2022.
- [34] Y. Bokobza, R. Dabora, and K. Cohen, "Deep reinforcement learning for simultaneous sensing and channel access in cognitive networks," *IEEE Transactions on Wireless Communications*, 2023.
- [35] R. Paul, K. Cohen, and G. Kedar, "Multi-flow transmission in wireless interference networks: A convergent graph learning approach," *IEEE Transactions on Wireless Communications*, pp. 1–1, 2023.
- [36] R. Adeogun and G. Berardinelli, "Distributed channel allocation for mobile 6g subnetworks via multi-agent deep q-learning," in *2023 IEEE Wireless Communications and Networking Conference (WCNC)*, pp. 1–6, IEEE, 2023.
- [37] A. Kaur, J. Thakur, M. Thakur, K. Kumar, A. Prakash, and R. Tripathi, "Deep recurrent reinforcement learning-based distributed dynamic spectrum access in multichannel wireless networks with imperfect feedback," *IEEE Transactions on Cognitive Communications and Networking*, vol. 9, no. 2, pp. 281–292, 2023.
- [38] S. Wang, H. Liu, P. H. Gomes, and B. Krishnamachari, "Deep reinforcement learning for dynamic multichannel access in wireless networks," *IEEE Transactions on Cognitive Communications and Networking*, vol. 4, no. 2, pp. 257–265, 2018.
- [39] V. Mnih, K. Kavukcuoglu, D. Silver, A. Graves, I. Antonoglou, D. Wierstra, and M. Riedmiller, "Playing atari with deep reinforcement learning," *arXiv preprint arXiv:1312.5602*, 2013.
- [40] M. Hessel, J. Modayil, H. Van Hasselt, T. Schaul, G. Ostrovski, W. Dabney, D. Horgan, B. Piot, M. Azar, and D. Silver, "Rainbow: Combining improvements in deep reinforcement learning," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 32, 2018.
- [41] J. Schrittwieser, I. Antonoglou, T. Hubert, K. Simonyan, L. Sifre, S. Schmitt, A. Guez, E. Lockhart, D. Hassabis, T. Graepel, *et al.*, "Mastering atari, go, chess and shogi by planning with a learned model," *Nature*, vol. 588, no. 7839, pp. 604–609, 2020.
- [42] O. Vinyals, I. Babuschkin, W. M. Czarnecki, M. Mathieu, A. Dudzik, J. Chung, D. H. Choi, R. Powell, T. Ewalds, P. Georgiev, *et al.*, "Grandmaster level in starcraft ii using multi-agent reinforcement learning," *Nature*, vol. 575, no. 7782, pp. 350–354, 2019.
- [43] Y. Li, "Deep reinforcement learning: An overview," *arXiv preprint arXiv:1701.07274*, 2017.
- [44] Y. Xing, R. Chandramouli, S. Mangold, and N. SS, "Dynamic spectrum access in open spectrum wireless networks," *IEEE journal on selected areas in communications*, vol. 24, no. 3, pp. 626–637, 2006.
- [45] H. Van Hasselt, A. Guez, and D. Silver, "Deep reinforcement learning with double q-learning," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 30, 2016.
- [46] A. Mishra, V. Shrivastava, S. Banerjee, and W. Arbaugh, "Partially overlapped channels not considered harmful," in *Proceedings of the joint international conference on Measurement and modeling of computer systems*, pp. 63–74, 2006.
- [47] X. Zhao, L. Li, S. Geng, H. Zhang, and Y. Ma, "A link-based variable probability learning approach for partially overlapping channels assignment on multi-radio multi-channel wireless mesh information-centric iot networks," *IEEE Access*, vol. 7, pp. 45137–45145, 2019.
- [48] M. T. Nguyen and L. B. Le, "Interference cancellation, channel estimation, and symbol detection for communications on overlapping channels," *IEEE Access*, vol. 8, pp. 89822–89838, 2020.
- [49] S. Kim, K. Asadi, M. Littman, and G. Konidaris, "Deepmellow: removing the need for a target network in deep q-learning," in *Proceedings of the twenty eighth international joint conference on artificial intelligence*, 2019.
- [50] C. Huang, V. J. Sorger, M. Miscuglio, M. Al-Qadasi, A. Mukherjee, L. Lampe, M. Nichols, A. N. Tait, T. Ferreira de Lima, B. A. Marquez, *et al.*, "Prospects and applications of photonic neural networks," *Advances in Physics: X*, vol. 7, no. 1, p. 1981155, 2022.
- [51] J. J. Egli, "Radio propagation above 40 mc over irregular terrain," *Proceedings of the IRE*, vol. 45, no. 10, pp. 1383–1391, 1957.
- [52] Y. Arjoune and N. Kaabouch, "A comprehensive survey on spectrum sensing in cognitive radio networks: Recent advances, new challenges, and future research directions," *Sensors*, vol. 19, no. 1, p. 126, 2019.
- [53] A. Nasser, H. Al Haj Hassan, J. Abou Chaaya, A. Mansour, and K.-C. Yao, "Spectrum sensing for cognitive radio: Recent advances and future challenge," *Sensors*, vol. 21, no. 7, p. 2408, 2021.
- [54] R. S. Sutton and A. G. Barto, *Reinforcement learning: An introduction*. MIT press, 2018.
- [55] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski, *et al.*, "Human-level control through deep reinforcement learning," *nature*, vol. 518, no. 7540, pp. 529–533, 2015.
- [56] A. Feriani and E. Hossain, "Single and multi-agent deep reinforcement learning for ai-enabled wireless networks: A tutorial," *IEEE Communications Surveys & Tutorials*, vol. 23, no. 2, pp. 1226–1252, 2021.
- [57] L. Canese, G. C. Cardarilli, L. Di Nunzio, R. Fazzolari, D. Giardino, M. Re, and S. Spanò, "Multi-agent reinforcement learning: A review of challenges and applications," *Applied Sciences*, vol. 11, no. 11, p. 4948, 2021.
- [58] P. Auer, N. Cesa-Bianchi, Y. Freund, and R. E. Schapire, "The non-stochastic multiarmed bandit problem," *SIAM journal on computing*, vol. 32, no. 1, pp. 48–77, 2002.
- [59] K. Asadi and M. L. Littman, "An alternative softmax operator for reinforcement learning," in *International Conference on Machine Learning*, pp. 243–252, PMLR, 2017.
- [60] X. Glorot and Y. Bengio, "Understanding the difficulty of training deep feedforward neural networks," in *Proceedings of the thirteenth international conference on artificial intelligence and statistics*, pp. 249–256, JMLR Workshop and Conference Proceedings, 2010.
- [61] A. Goldsmith, *Wireless communications*. Cambridge university press, 2005.
- [62] O. Vinyals, T. Ewalds, S. Bartunov, P. Georgiev, A. S. Vezhnevets, M. Yeo, A. Makhzani, H. Küttler, J. Agapiou, J. Schrittwieser, *et al.*, "Starcraft ii: A new challenge for reinforcement learning," *arXiv preprint arXiv:1708.04782*, 2017.
- [63] C. Berner, G. Brockman, B. Chan, V. Cheung, P. Dębiak, C. Dennison, D. Farhi, Q. Fischer, S. Hashme, C. Hesse, *et al.*, "Dota 2 with large scale deep reinforcement learning," *arXiv preprint arXiv:1912.06680*, 2019.
- [64] D. Ye, Z. Liu, M. Sun, B. Shi, P. Zhao, H. Wu, H. Yu, S. Yang, X. Wu, Q. Guo, *et al.*, "Mastering complex control in moba games with deep reinforcement learning," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 34, pp. 6672–6679, 2020.
- [65] Y. Hou, X. Liang, J. Zhang, Q. Yang, A. Yang, and N. Wang, "Exploring the use of invalid action masking in reinforcement learning: A comparative study of on-policy and off-policy algorithms in real-time strategy games," *Applied Sciences*, vol. 13, no. 14, p. 8283, 2023.
- [66] C. Blum and A. Roli, "Metaheuristics in combinatorial optimization: Overview and conceptual comparison," *ACM computing surveys (CSUR)*, vol. 35, no. 3, pp. 268–308, 2003.